NutriAI

Personalised 7-Day Diet Planning powered by ML

Final Project | Option A | June 05, 2026

EXECUTIVE SUMMARY

NutriAI is a production-grade automated diet planning application that generates a personalised 7-day meal plan in 2-3 seconds, strictly tailored to a user's clinical conditions, dietary preferences, allergen restrictions, and daily nutritional targets. The system integrates all five BAX-423 ML tech

DATA PIPELINE & STATISTICS

Pipeline Stages

1. USDA Ingest	build_database.py - Foundation + SR Legacy + FNDDS
2. Enrichment	enrich_database.py - clinical tags, allergen flags, GI estimates
3. Clinical Filter	SQL WHERE: FODMAP, GERD, GI<=55, sodium cap, 50+ name guards
4. ANN Retrieval	15-dim nutrient embeddings, cosine k-NN (k=120) via sklearn
5. Bloom Filter	Belt-and-suspenders allergen check, 0% false-negative rate
6. Multi-obj Rank	5-dimension weighted scorer (calorie, gap-fill, diversity, priority, clinical)
7. Streaming Output	Python generator yields each meal live; Streamlit updates in real time
8. Gap Feedback	Day totals feed back into gap vector; next slot adapts to what

Key Statistics

Foods in DB:	13,620
Nutrient columns:	21 per food
Data types:	Foundation, SR Legacy, FNDDS
Allergen filters:	11 types, Bloom-backed
Clinical conditions:	IBS, GERD, T2DM, Hypertension
Diet modes:	4 (vegan, vegetarian, pescat., omni)
BAX-423 techniques:	5 of 5 applicable
Generation time:	2-3 s end-to-end
Diversity index:	0.78-0.87 (Simpson's D)
Personas passing:	4 / 4 (all 6 checks)

SIGNATURE DELIVERABLES

"Why Excluded":	Every filtered food returns a clinical/allergen reason string (explainer.py)
PDF + CSV Export:	10-page PDF plan | 3 CSVs (meals, daily summary, exclusions) + ZIP
Sub-60s Generation:	Typical cold start ~2-3s; warm (cached Bloom+embeddings) <1s
Pass/Fail Table:	All 4 personas x 6 capabilities - see Page 3

BAX-423 TECHNIQUES & BENCHMARKS

T1 - SKETCHING (Bloom Filter)

Probabilistic Data Structures for allergen safety

11 AllergenFilterBank instances (one per allergen type). Each filter: m = 1.2M bits, k = 7 hash functions (MD5 + SHA-256 two-hash trick). Built once at session start from the full 13,620-food database. Applied AFTER SQL allergen clauses - belt-and-suspenders: SQL handles the bulk exclusion; Bloom gu

Metric	
Build time (13,620 foods, 11 filters)	**873 ms (one-time per session)**
Lookup time per food	**97.8 mcgs (O(1), k=7 hashes)**
False-negative rate	**0.000% (by construction)**
False-positive rate	**~0.08% (harmless: only over-excludes)**
vs SQL-only safety	**SQL alone: ~0.1% allergen escape on complex composites**

T2 - EMBEDDINGS (Nutrient Vectors)

Vector Representations for nutritional similarity

15-dimensional nutrient embedding per food: [calories, protein, carbs, fat, fiber, calcium, iron, potassium, magnesium, zinc, B12, vit_D, vit_C, omega3_ALA, omega3_EPA]. Transform pipeline: weighted (B12=2.0, vit_D=1.8, iron=1.5 to amplify scarce nutrients) -> log1p (compress concentration outliers)

Metric	
Index build (5,000 foods)	**~60 ms**
Query time (k=120)	**<5 ms**
Gap-fill score: ANN top-5	**5.26 (nutrient relevance index)**
Gap-fill score: random 5 foods	**1.60 (baseline)**
ANN improvement over random	**+230% - retrieves foods 3.3x more aligned to gap**

T3 - RECOMMENDATION (Content-Based ANN)

Recommendation Systems - adaptive gap-filling content filter

Content-based filtering: the query IS the user's current nutritional state. After each meal is assigned, day_totals updates, compute_gap_vector() recalculates the fractional remaining RDA, and the next ANN query reflects what was actually consumed - creating a closed adaptive feedback loop across al

Metric	
Adaptive iterations per plan	**21 (gap vector recalculated after every meal)**
Candidates retrieved per slot	**120 (cosine ANN from candidate pool)**
Pool sizes (typical)	**2,600 (Priya) to 4,500 (Ravi)**
Coverage vs fixed-gap baseline	**Adaptive: 4/7 RDA days (Priya) vs 0/7 fixed**

T4 - RANKING (Multi-Objective Weighted Score)

Ranking & Learning-to-Rank - five-dimension weighted scorer

5 scoring dimensions combined as weighted sum -> final score [0,1]. W1=0.20 Calorie Fit: Gaussian on delivered_kcal/slot_target, sigma=0.5, with per-day budget guard (<=110% of daily target) and per-slot calorie density cap. W2=0.32 Nutrient Gap-Fill: fractional RDA fill weighted by relative gap urg

Metric	
Diversity - multi-objective ranker	**0.816 (Simpson's D, Priya; range 0.78-0.87)**
Diversity - random selection baseline	**0.617 (same food pool, no ranking)**
Diversity improvement	**+0.199 (+32% better food-group spread)**
Calorie adherence (all personas)	**0/7 days exceed 110% of target (after budget guard)**
RDA coverage: Priya "days met"	**4/7 days (vs 0/7 before gap-fill reweighting)**

T5 - STREAMING (Progressive Plan Generation)

Streaming & Real-Time Processing - live meal-by-meal delivery

generate_plan_stream() is a Python generator. Each of 21 meal slots yields a typed event dict {type:"meal", day, slot, meal, elapsed_s} immediately after selection. Streamlit iterates the generator and updates the progress bar + live description for each event. The user sees meals appearing one-by-o

Metric	
Time to first meal (cold start)	**~2.1 s (includes Bloom build + embedding index)**
Total plan time (cold start)	**~3.0 s**
Time to first meal (warm, cached)	**<200 ms (Streamlit @st.cache_resource)**
Events yielded per plan	**22 (21 meals + 1 "complete" event)**
vs batch (wait for all 21)	**User sees plan building live; 17% faster first result**

PERSONA PASS / FAIL TABLE - All 4 Test Personas x 6 Core Capabilities

Priya (28y F, 1800 kcal)

[illegible] Weight / Protein floor: [illegible] g/day protein

Ravi (38y M, 2200 kcal)

Diet: Non-veg / No pork
Condition: GERD
Allergen: Gluten (Celiac)
Priority Micros: B12, Zinc, Mg
Weight / Protein floor: 80 kg => >56 g/day protein

Mei (45y F, 1600 kcal)

[illegible] Weight / Protein floor: [illegible] g/day protein

James (52y M, 2000 kcal)

Diet: Pescatarian
Condition: Hypertension
Allergen: Soy
Priority Micros: Potassium, Mg, Omega-3
Weight / Protein floor: 88 kg => >62 g/day protein

Capability	Priya	Ravi	Mei	James
C1 Clinical Condition Filtering	PASS	PASS	PASS	PASS
FODMAP, low-acid, GI<=55, DASH sodium cap				
C2 Allergy Detection & Exclusion	PASS	PASS	PASS	PASS
SQL + Bloom filter; zero false negatives				
C3 Dietary Preference Handling	PASS	PASS	PASS	PASS
Vegetarian / vegan / pescatarian / non-veg				
C4 Diversity Engine	0.816	0.871	0.798	0.834
No food repeated; Simpson's D >= 0.70				
C5 Macro & Micro Analysis	PASS	PASS	PASS	PASS
21 nutrients tracked; RDA gaps flagged at 80%				
C6 Sub-60s Generation	3.0s	2.9s	2.1s	2.1s
Cold ~2s; warm (cached) <1s				

KEY OUTCOME METRICS (per persona)

Metric	Priya	Ravi	Mei	James
Generation time	3.0 s	2.9 s	2.1 s	2.1 s
Candidate pool size	2,420	4,232	2,269	2,790
Diversity (Simpson's D)	0.839	0.871	0.798	0.834
Days >= 80% RDA (+)	4 / 7	0 / 7 *	1 / 7 *	0 / 7 *
Calorie adherence	Avg 95%	Avg 97%	Avg 91%	Avg 94%
Days >110% calories	0 / 7	0 / 7	0 / 7	0 / 7
Iron >= 80% RDA daily	PASS 6/7	-	-	-
Fiber >= 25g/day	-	-	3-5 / 7	-
Potassium >= 80% RDA	-	-	-	4 / 7
Sodium <= 1500mg/day	-	-	-	2 / 7 **

*(+) "Days >= 80% RDA" excludes Vitamin D (primarily sunlight-dependent; food alone <5% of 600 IU RDA), Omega-3 ALA (plant sources excluded as non-standalone-meal items), and Sodium (a cap, not a floor). * Structural constraints: Ravi fiber 38g/day near-impossible with GERD+gluten-free; Mei calories/fat on vegan+T2DM+1600 kcal; James fat/fiber on pescatarian+hypertension+soy-free. ** James sodium: cumulative multi-slot overshoot; individual foods all pass is_high_sodium=0 filter.*

ADAPTIVE LEARNING - CLOSED-LOOP GAP FEEDBACK

NutriAI implements within-plan adaptive learning: after each meal is assigned, compute_day_totals() aggregates actual nutrients delivered, and compute_gap_vector() recalculates fractional remaining RDA needs (Eq: gap[n] = max(0, (RDA - actual) / RDA)). The ANN query and the ranker's gap-fill score b

DEPLOYMENT & SUBMISSION

Live App URL:	https://nutriai-bax423-msba.streamlit.app (Streamlit Community Cloud)
Tech Stack:	Python 3.13, Streamlit, SQLite, scikit-learn, fpdf2, plotly, bitarray
Database:	13,620 foods | nutriai_foods.db | ~10 MB SQLite | in repo
Setup:	pip install -r requirements.txt -> streamlit run app.py
Rebuild DB:	python build_database.py (13 min, USDA API) + python enrich_database.py (24s)
Tests:	python test_personas.py - all 4 personas, 6 capability checks, ~5 s
ZIP Contents:	code/ (full src), data/ (nutriai_foods.db snapshot), brief.pdf, prompts.md, README.md

KNOWN LIMITATIONS (Honest Assessment)

Ravi 0/7 RDA days:

Fiber RDA of 38 g/day (male) is clinically near-impossible with GERD + gluten-free combined. The pipeline correctly avoids bran/psyllium (excluded as supplements) and acidic high-fiber foods (GERD triggers). This reflects real dietary constraint, not a pipeline failure.

James cumulative sodium:

The ranker excludes individually high-sodium foods (>400 mg/100 g) via is_high_sodium=0 SQL flag, but three moderate-sodium foods combined can exceed the 1,500 mg/day DASH cap. Fix: pass rolling day sodium into the gap vector as a soft penalty once budget > 60%.

Vitamin D & Omega-3 ALA:

Food-only Vitamin D achieves <5% of the 600 IU RDA for dairy-free profiles. Omega-3 ALA requires nuts/seeds/oils which are excluded as non-standalone meal items. Both are excluded from the "days met" UI metric with a footnote; supplementation is recommended.

Calorie variation +/-10%:

Daily calorie delivery varies +/-5-10% around the user target. A per-slot calorie density cap (scaled to user calorie target) and a per-day budget guard (<=110%) prevent large outliers. Average adherence is 91-97% across all personas. Exact daily calorie matching would require further narrowing the

No user feedback persistence:

Adaptive learning operates within a single plan generation. User preferences (liked/disliked foods) are not persisted across sessions. A future version could store ratings in a local JSON and use them to bias food group weights in subsequent generations.

TECHNICAL REFLECTION

The two most impactful design decisions were (1) the fractional gap-vector normalisation and (2) the calorie-proportional serving size. Before normalisation, absolute units caused calcium (1,000 mg RDA) to dominate fiber (25 g) by 40x in the gap-fill score, making the ANN query and ranker effectivel